Chaoyi Lu

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Education & Employment

2025 - Now: Associate research fellow — **Zhongguancun Laboratory**

2022 - 2025: Postdoctoral researcher — **Tsinghua University**

Reporting to Prof. Haixin Duan

2017 - 2022: PhD — **Tsinghua University**

Advisor: Prof. Jianping Wu

Dissertation: *On Measuring Security Technologies within the Domain Name System Hierarchy*

2013 - 2017: BSc — **Beijing University of Posts and Telecommunications**

Research & Publications

- **28 peer-reviewed academic papers + 1 monograph + 1 industrial standard + 2 patents**
  - *20 at top-tier venues: CCS (x5), NDSS (x5), USENIX Security (x4), IMC (x4), WWW (x2)*
- **Research topics:**
  - *Protocol operations measurement*: *encrypted DNS* [10, 22], *DNS servicing* [17, 21, 23, 31]
  - *Large-scale data analysis*: *privacy* [5, 15], *traffic manipulation* [8-9, 24, 30], *UA* [7]
  - *New security threats*: *denial-of-service* [19-20, 27], *DNS cache poisoning* [11, 18, 28, 32]
  - *Other infrastructure*: *Web PKI* [13, 16], *Email* [14, 25-26], *Mobile* [12]
- **Awards:**
  - *IRTF/ISOC Applied Networking Research Prize*
  - *ACM CCS Distinguished Paper Award, NDSS Distinguished Paper Award*

Peer-reviewed papers

[32] **[CCS '26]** S. Liu, Y. Zhang, **Chaoyi Lu**, B. Liu, S. Zhang, et al. ***Good Cache, BAD Cache: Exploiting DNSSEC Validation Failures for DNS Cache Poisoning Attacks.*** In Proceedings of the 33rd ACM Conference on Computer and Communications Security, November 2026 *(To appear)*

[31] **[IMC '26]** Y. Zhang, H. Zhou, T. Wan, H. Duan, L. Hao, et al. ***Unintended Revelations and Risks: Understanding Cellular DNS Leakage on the Public Internet.*** In Proceedings of the 26th ACM Internet Measurement Conference, October 2026.

[30] **[ToN vol.34]** Y. Xing, **Chaoyi Lu**, B. Liu, R. Li, H. Duan. ***Traffic Shadowing: A Global Investigation of Internet Traffic Observation and User Data Reutilization.*** In IEEE Transactions on Networking, vol. 34, June 2026

[29] **[WWW '26]** S. Cui, M. Liu, B. Liu, H. Duan, R. Li, **Chaoyi Lu**, et al. ***Characterizing Iran's Phased National Internet Shutdown in 2025: A Progressive and Distributed Action.*** In Proceedings of the 38th ACM Web Conference, June 2026.

[28] **[NDSS '26]** Y. Wu, Y. Zhang, **Chaoyi Lu**, B. Liu. ***Should I Trust You? Rethinking the Principle of Zone-Based Isolation DNS Bailiwick Checking.*** In Proceedings of the 33rd Network and Distributed System Security Symposium, February 2026.

◇ **Recipient of NDSS Distinguished Paper Award**

[27] **[NDSS '26]** R. Li, **Chaoyi Lu**, B. Liu, Y. Lin, Q. Pan, et al. ***CoordMail: Exploiting SMTP Timeout and Command Interaction to Coordinate Email Middleware for Convergence Amplification Attack.*** In Proceedings of the 33rd Network and Distributed System Security Symposium, February 2026.

[26] **[IMC '25]** R. Li, **Chaoyi Lu**, B. Liu, Y. Lin, H. Duan, et al. ***Understanding and Characterizing Intermediate Paths of Email Delivery: The Hidden Dependencies.*** In Proceedings of the 25th Internet Measurement Conference, October 2025.

[25] **[NDSS '25]** R. Li, **Chaoyi Lu**, B. Liu, Y. Zhang, G. Hong, et al. ***HADES Attack: Understanding and Evaluating Manipulation Risks of Email Blocklists.*** In Proceedings of the 32nd Network and Distributed System Security Symposium, February 2025.

[24] **[IMC '24]** Y. Xing*, **Chaoyi Lu***, B. Liu, H. Duan, J. Sun, et al. ***Yesterday Once More: Global Measurement of Internet Traffic Shadowing Behaviors.*** In Proceedings of the 24th Internet Measurement Conference, November 2024. *(*Co-first authors)*

[23] **[TDSC vol.21]** F. Zhang, B. Liu, **Chaoyi Lu***, Y. Xing, H. Duan, et al. ***Investigating Deployment Issues of DNS Root Server Instances From a China-Wide View.*** In IEEE Transactions on Dependable and Secure Computing, Volume 21, Issue 6, November 2024.

[22] **[WWW '24]** R. Li, B. Liu, **Chaoyi Lu**, H. Duan, J. Shao. ***A Worldwide View on the Reachability of Encrypted DNS Services.*** In Proceedings of the 36th Web Conference, May 2024.

[21] **[NDSS '24]** M. Liu, Y. Zhang, X. Li, **Chaoyi Lu**, B. Liu, et al. ***Understanding the Implementation and Security Implications of Protective DNS Services.*** In Proceedings of the 31st Network and Distributed System Security Symposium, February 2024.
◇ Community event contribution: ICANN APAC DNS Forum (Pre-event webinar) 2024

[20] **[CCS '23a]** W. Xu, X. Li, **Chaoyi Lu**, B. Liu, J. Zhang, et al. ***TsuKing: Coordinating DNS Resolvers and Queries into Potent DoS Amplifiers.*** In Proceedings of the 30th ACM Conference on Computer and Communications Security, November 2023.
◇ Community event contribution: Black Hat Europe 2023, OARC 41

[19] **[CCS '23b]** F. Zhang, B. Liu, E. Alowaisheq, J. Chen, **Chaoyi Lu**, et al. ***Silence is not Golden: Disrupting the Load Balancing of Authoritative DNS Servers.*** In Proceedings of the 30th ACM Conference on Computer and Communications Security, November 2023.
◇ Recipient of ACM CCS Distinguished Paper Award
◇ Community event contribution: OARC 41, ICANN APAC DNS Forum (Pre-event webinar) 2023

[18] **[USENIX Sec '23]** X. Li, **Chaoyi Lu**, B. Liu, Q. Zhang, Z. Li, et al. ***The Maginot Line: Attacking the Boundary of DNS Caching Protection.*** In Proceedings of the 32nd USENIX Security Symposium, Anaheim, CA, USA, August 2023.

[17] **[PAM '22]** F. Zhang, **Chaoyi Lu**, B. Liu, H. Duan, Y. Liu. ***Measuring the Practical Effect of DNS Root Server Instances: A China-Wide Case Study.*** In Proceedings of the 23rd Passive and Active Measurement Conference. March 2022.
◇ Community event contribution: ICANN APAC DNS Forum 2022

[16] **[CCS '21]** Y. Zhang, B. Liu, **Chaoyi Lu**, Z. Li, H. Duan, et al. ***Rusted Anchors: A National Client-Side View of Hidden Root CAs in the Web PKI Ecosystem.*** In Proceedings of the 28th ACM Conference on Computer and Communications Security, November 2021.

[15] **[NDSS '21]** **Chaoyi Lu**, B. Liu, Y. Zhang, Z. Li, F. Zhang, H. Duan, et al. ***From WHOIS to WHOWAS: A Large-Scale Measurement Study of Domain Registration Privacy under the GDPR.*** In Proceedings of the 28th Network and Distributed System Security Symposium, February 2021.

[14] **[USENIX Sec '21]** K. Shen, C. Wang, X. Zheng, M. Guo, **Chaoyi Lu**, et al. ***Weak Links in Authentication Chains: A Large-scale Analysis of Email Sender Spoofing Attacks.*** In Proceedings of the 30th USENIX Security Symposium, August 2021.

[13] **[CCS '20a]** M. Zhang, X. Zheng, K. Shen, Z. Kong, **Chaoyi Lu**, et al. ***Talking with Familiar Strangers: An Empirical Study on HTTPS Context Confusion Attacks.*** In Proceedings of the 27th ACM Conference on Computer and Communications Security, November 2020.

[12] **[CCS '20b]** Y. Zhang, B. Liu, **Chaoyi Lu**, Z. Li, H. Duan, et al. ***Lies in the Air: Characterizing Fake-base-station Spam Ecosystem in China.*** In Proceedings of the 27th ACM Conference on Computer and Communications Security, November 2020.

[11] **[USENIX Sec '20]** X. Zheng, **Chaoyi Lu**, J. Peng, Q. Yang, D. Zhou, et al. ***Poison Over Troubled Forwarders: A Cache Poisoning Attack Targeting DNS Forwarding Devices.*** In Proceedings of the 29th USENIX Security Symposium, August 2020.

◇ Community event contribution: ICANN DNS Symposium 2021

[10] **[IMC '19]** **Chaoyi Lu**, B. Liu, Z. Li, S. Hao, H. Duan, et al. ***An End-to-End, Large-Scale Measurement of DNS-over-Encryption: How Far Have We Come?.*** In Proceedings of the 19th Internet Measurement Conference, October 2019.

◇ Recipient of IRTF/ISOC Applied Networking Research Prize (ANRP)
◇ Nominee of Distinguished Paper Award & Community Contribution Award
◇ Community event contribution: IETF 108, OARC 31

[9] **[EuroS&P '19]** B. Liu, Z. Li, P. Zong, **Chaoyi Lu**, H. Duan, et al. ***TraffickStop: Detecting and Measuring Illicit Traffic Monetization Through Large-scale DNS Analysis.*** In Proceedings of the 4th IEEE European Symposium on Security and Privacy, June 2019.

[8] **[USENIX Sec '18]** B. Liu, **Chaoyi Lu**, H. Duan, Y. Liu, Z. Li, et al. ***Who Is Answering My Queries: Understanding and Characterizing Interception of the DNS Resolution Path.*** In Proceedings of the 27th USENIX Security Symposium, August 2018.

◇ Community event contribution: ACM/IRTF ANRW '19, OARC 30

[7] **[DSN '18]** B. Liu, **Chaoyi Lu**, Z. Li, Y. Liu, H. Duan, et al. ***A Reexamination of Internationalized Domain Names: the Good, the Bad and the Ugly.*** In Proceedings of the 48th IEEE/IFIP International Conference on Dependable Systems and Networks, June 2018.

[6] **[EthiCS Workshop '22]** Y. Zhang, M. Liu, M. Zhang, **Chaoyi Lu**, H. Duan. ***Ethics in Security Research: Visions, Reality, and Paths Forward.*** 1st International Workshop on Ethics in Computer Security. June 2022.

◇ Recipient of Best Student Paper Award

[5] **[FOCI Workshop '18]** M. Zhang, B. Liu, **Chaoyi Lu**, J. Zhang, S. Hao, H. Duan. ***Measuring Privacy Threats in China-Wide Mobile Networks.*** 8th USENIX Workshop on Free and Open Communications on the Internet, August 2018.

Monograph

[4] **Chaoyi Lu**. (2024). ***On Measuring Security Technologies within the Domain Name System Hierarchy.*** Tsinghua University Press. *(Monograph written in Chinese)*

Industrial standard & patents

[3] **Chaoyi Lu**, Y. Huang, J. Zhang, H. Duan, D. Ma, et al. ***Domain Name System Resolution Data Encryption Transmission Technical Requirements.*** Industrial standard of telecommunication technologies of the P.R.C. *(No. YD/T 4712-2024, effective Jul 2024)*

[2] J. Li, Y. Zhang, B. Liu, **Chaoyi Lu**, H. Duan, H. Wang. ***Server Security Inspection Method and Apparatus Based on Domain Name List.*** China Patent CN116545705B, authorized 2025-12-02.

[1] X. Li, **Chaoyi Lu**, B. Liu, H. Duan, Q. Li. ***Domain name resolver identification method, apparatus and electronic device***. China Patent CN116708361B, authorized 2026-02-27.

Services

Internet community committee member

ICANN Security and Stability Advisory Committee (SSAC), 2024-Present

ICANN SSAC DNSSEC Operational Considerations Work Party Co-chair, 2025-Present

Conference TPC member

WWW '27, EAI SecureComm '23

External reviewer

Conferences: RAID '23, NDSS '22 & '20, ICDCS '21, DSN '20, CCS '19, ESORICS '19, ICICS '19

Journal & Workshops: IEEE TNSM 2020, WPES '21, IoT S&P '18, RESEC '18

Teaching & Events

Community events

Pre-event webinar moderator, ICANN APAC DNS Forum, 2024

Plenary & pre-event webinar moderator, ICANN APAC DNS Forum, 2023

Technical training sessions

DNSSEC training for CERNET (under support of APNIC ISIF Project), 2023

Cybersecurity Fundamentals, Training for Malaysia Cyber 100 Initiative, 2023

Domain Name System Fundamentals, DataCon Student Summer Camp, 2022-2024

Courses at Tsinghua University

Invited talk, Basics of Computer Knowledge and Cultures, 2024 Fall

Invited talk, Internet Measurement and Security Analysis, 2023 Fall

Joint lecturer, Computer Network Security Experiments, 2023 Summer

Teaching assistant, Basics and Cutting-Edge Topics of Cyberspace, 2020 Fall

Teaching assistant, Computer Networks, 2019 Fall

Teaching assistant, Computer Network Security Technologies, 2019 Spring

Awards

Academia & community

ACM SIGSAC China "Rising Star" Award	2026
NDSS Distinguished Paper Award	2026
ACM CCS Distinguished Paper Award	2023
IRTF Applied Networking Research Prize (ANRP)	2020
ACM IMC Nominee of Distinguished Paper Award & Community Contribution Award	2019
IEEE/IFIP DSN Student Travel Grant	2018

Education & scholarship

Graduate Award, Education Commission of Beijing Municipality	2022 & 2017
PhD Graduate Award of Tsinghua University	2022
PhD Dissertation Award of Tsinghua University	2022
1st-Class Scholarship of Tsinghua University for Graduate Students	2021 & 2020
Cyberspace Scholarship of China Internet Development Foundation	2016